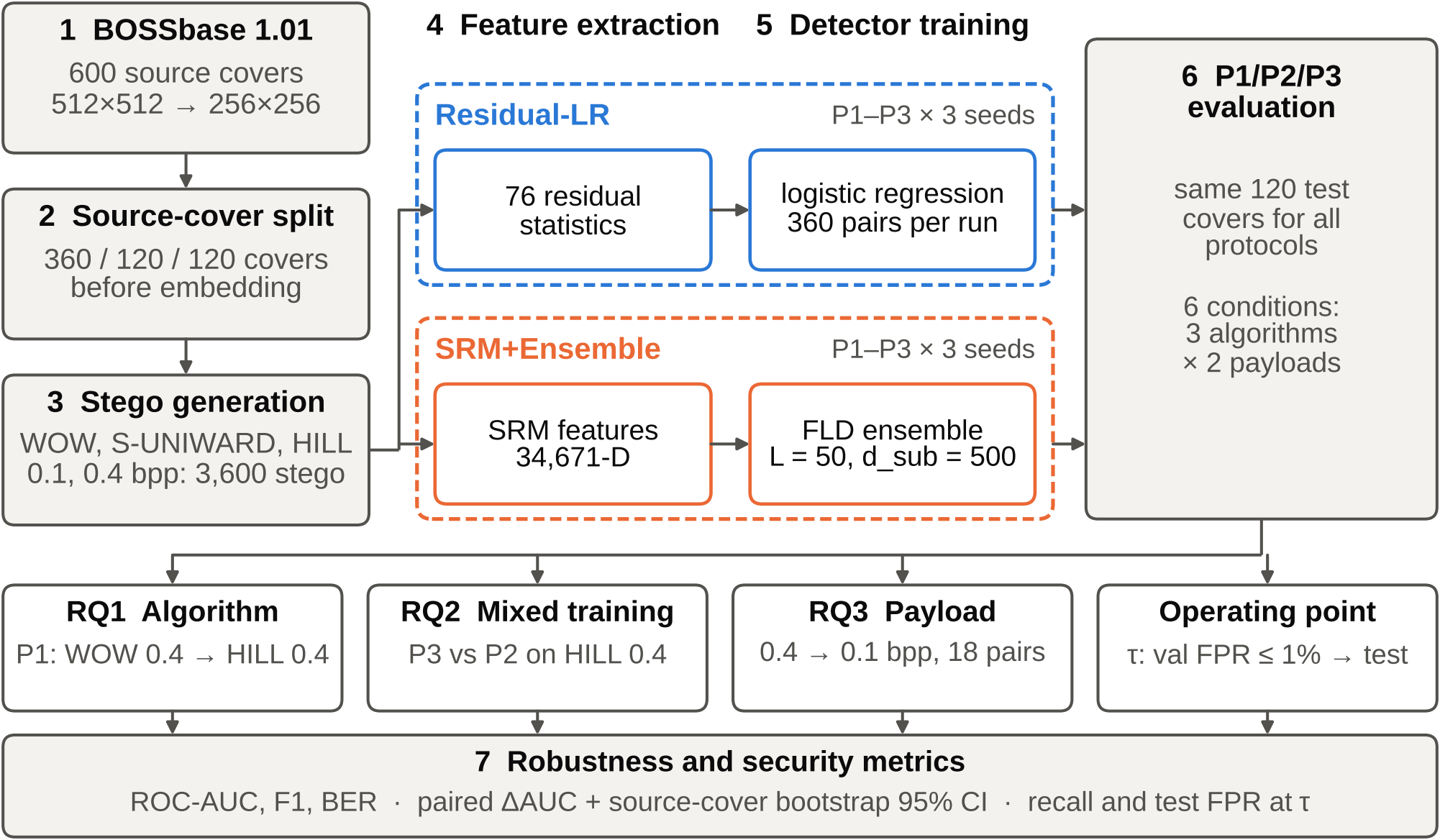
1 BOSSbase 1.01
600 source covers
512×512 → 256×256
2 Source-cover split
360 / 120 / 120 covers
before embedding
3 Stego generation
WOW, S-UNIWARD, HILL
0.1, 0.4 bpp: 3,600 stego
4 Feature extraction
5 Detector training
Residual-LR
P1–P3 × 3 seeds
76 residual
statistics
logistic regression
360 pairs per run
SRM+Ensemble
P1–P3 × 3 seeds
SRM features
34,671-D
FLD ensemble
L = 50, d_sub = 500
6 P1/P2/P3
evaluation
same 120 test
covers for all
protocols
6 conditions:
3 algorithms
× 2 payloads
RQ1 Algorithm
P1: WOW 0.4 → HILL 0.4
RQ2 Mixed training
P3 vs P2 on HILL 0.4
RQ3 Payload
0.4 → 0.1 bpp, 18 pairs
Operating point
τ: val FPR ≤ 1% → test
7 Robustness and security metrics
ROC-AUC, F1, BER · paired ΔAUC + source-cover bootstrap 95% CI · recall and test FPR at τ